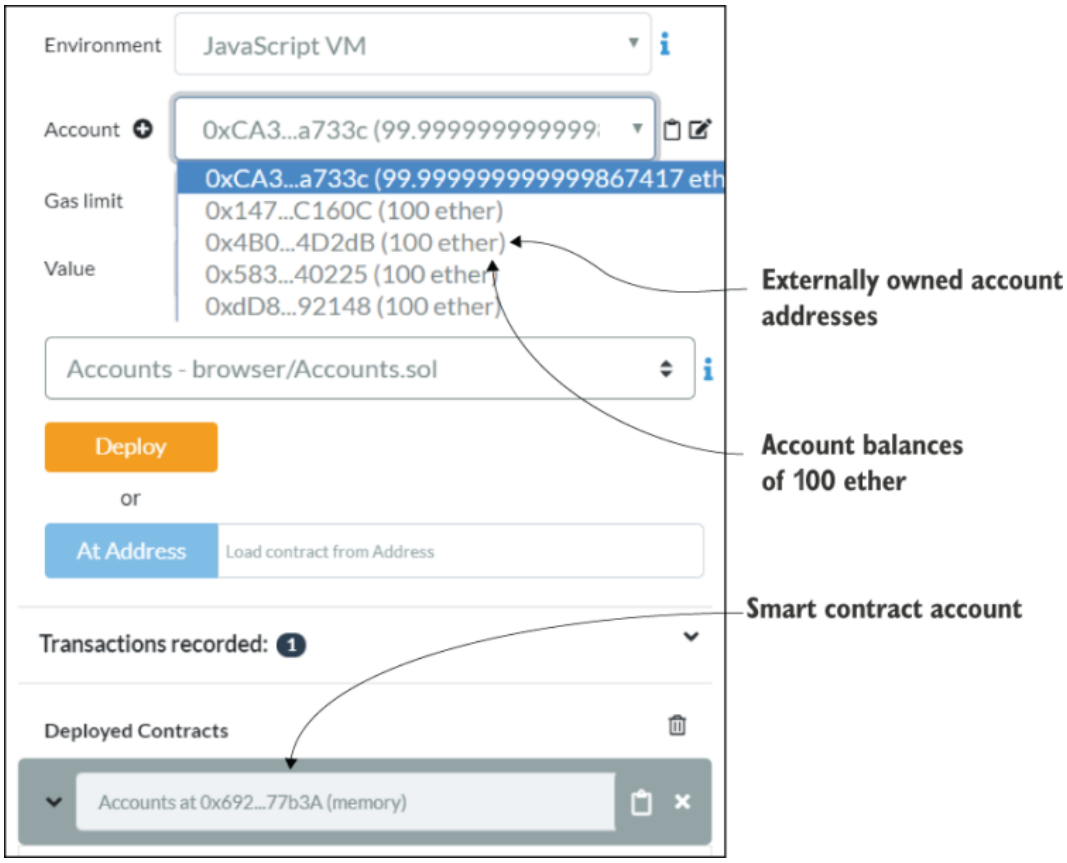


Figure 5.4 EOAs and smart contract accounts

Have you ever wondered how these account addresses (identities) are created? How is it that they are unique for the participants in the chain? To address these concerns, Ethereum uses a mechanism based in public-private key pairs to generate account addresses. Here is a high-level description of the mechanism:

1. A 256-bit random number is generated and designated as the private key.
2. A special algorithm called the elliptic curve cryptography algorithm is applied to this private key to derive a unique public key.

   These two form the `{private, public}` key pair; the private key is secured by a password, and the public key is open to the world.